Experiment-14

Aim

To perform image transformation using a Homography Matrix in Python with OpenCV by mapping one plane of an image onto another using corresponding points.

Procedure

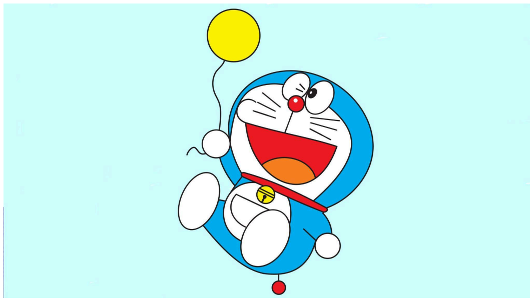
1. Import the required libraries (cv2 and numpy).
2. Load the input image.
3. Define four corresponding source points and destination points.
4. Compute the Homography Matrix using cv2.findHomography().
5. Apply the homography transformation using cv2.warpPerspective().
6. Display the original image and the transformed image.
7. Save the transformed image if required.

Code:

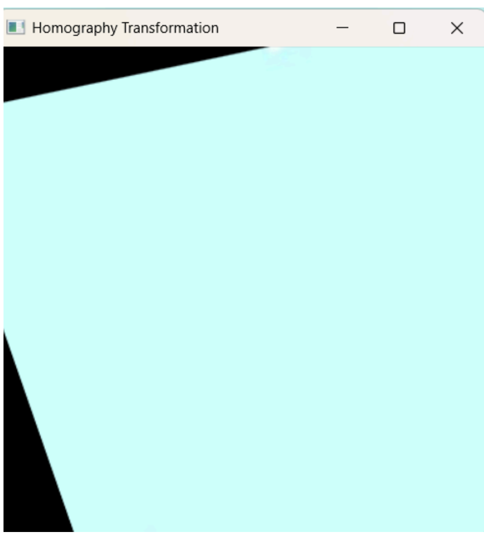
```
import cv2

import numpy as np

image = cv2.imread("sample.jpg")

if image is None:

    print("Error: Image not found!")

    exit()

src_pts = np.array([

    [50, 50],

    [300, 50],
```

```
    [50, 300],
    [300, 300]
], dtype=np.float32)
dst_pts = np.array([
    [10, 100],
    [280, 50],
    [80, 320],
    [320, 300]
], dtype=np.float32)
H, status = cv2.findHomography(src_pts, dst_pts)
result = cv2.warpPerspective(image, H, (400, 400))
cv2.imshow("Original Image", image)
cv2.imshow("Homography Transformation", result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:



Output:



Results:

The homography matrix was successfully computed using the corresponding source and destination points. The image was transformed using cv2.warpPerspective(), producing the desired perspective-corrected output.